



ArcherySec: Enhanced Containerized Vulnerability Assessment & Scanning Platform



Multi-Scanner Orchestration, Automated Scheduling & Transparent Risk Prioritization

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PROBLEM STATEMENT

Traditional vulnerability management relies on isolated web and network scanners that produce inconsistent outputs. This causes duplicate findings, manual triage, fragmented reporting and uneven prioritization. Inconsistent deployment and limited access governance further reduce reliability. ArcherySec SaaS addresses these issues through centralized orchestration, secure multi-tenant access, automated scheduling and transparent risk analysis.

PROJECT AIM

To modernize ArcherySec into a reliable, containerized and governance-focused vulnerability management platform that unifies scanner execution, scheduling, secure access, result processing and structured reporting in alignment with SDG 9.

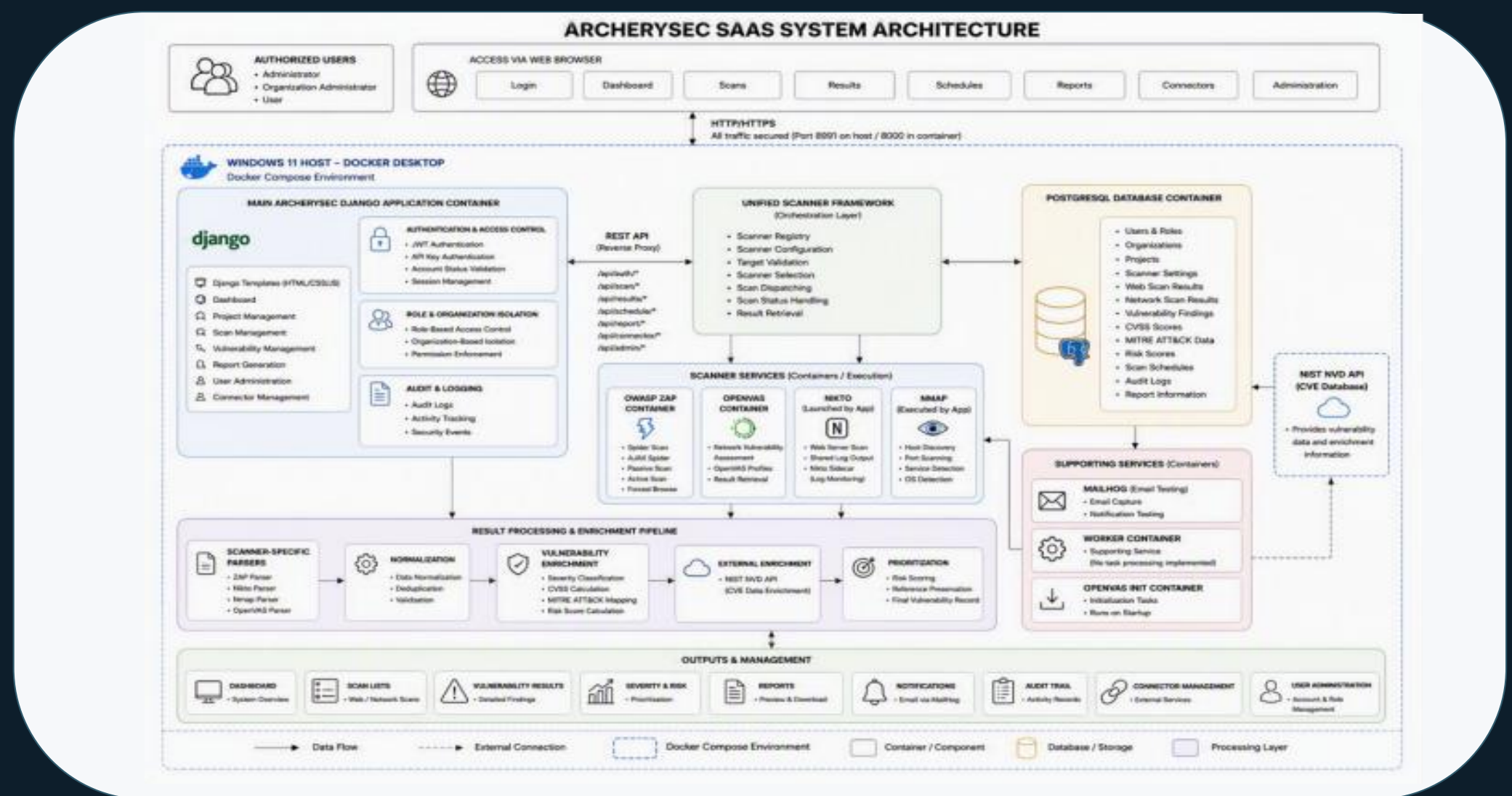
CONCLUSION

The platform centralizes web and network scanning, RBAC, enrichment, prioritization and reporting. All 25 functional checks passed, all 9,000 stress-test requests completed with no HTTP failures, and all three UAT participants accepted the system. The result was Pass with Performance Limitations because p95 reached 1.15 seconds at 50 virtual users.

OBJECTIVES

- Containerized Architecture: Reproducible Docker Compose deployment for the application, PostgreSQL and scanner services.
- Scanner Orchestration: Coordinate OWASP ZAP, Nikto, Nmap and OpenVAS through one workflow.
- Automation: Support configurable web and network scan scheduling.
- Governance: Apply JWT, API-key authentication, RBAC and organization isolation.
- Risk Analysis and Reporting: Apply CVSS, MITRE ATT&CK, NVD context and rule-based prioritization.

SYSTEM ARCHITECTURE



VERIFIED EVALUATION RESULTS

25 / 25
UNIT TESTS

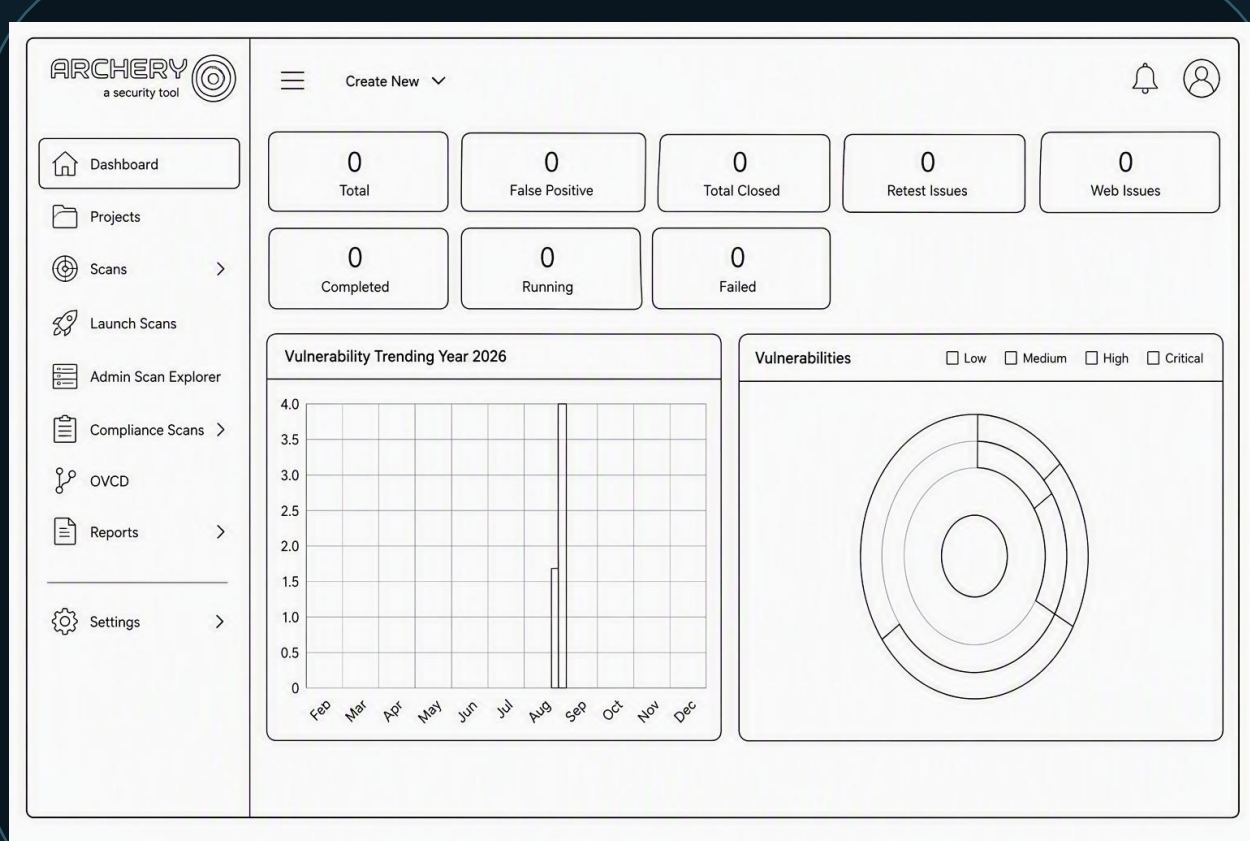
9,000
REQUESTS

0.00%
HTTP FAILURES

1.15 s
P95 AT 50 VUS

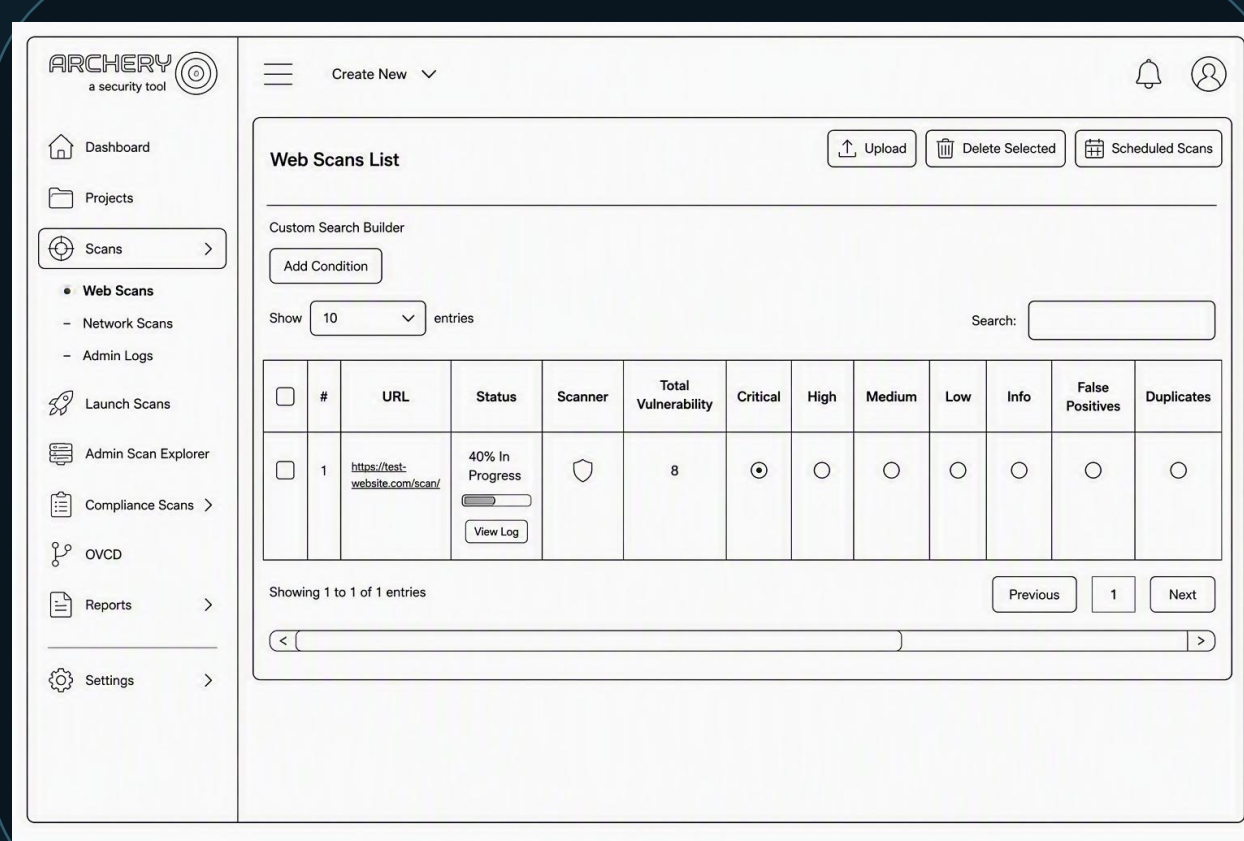
3 / 3
UAT ACCEPTED

SYSTEM INTERFACE & KEY FEATURES



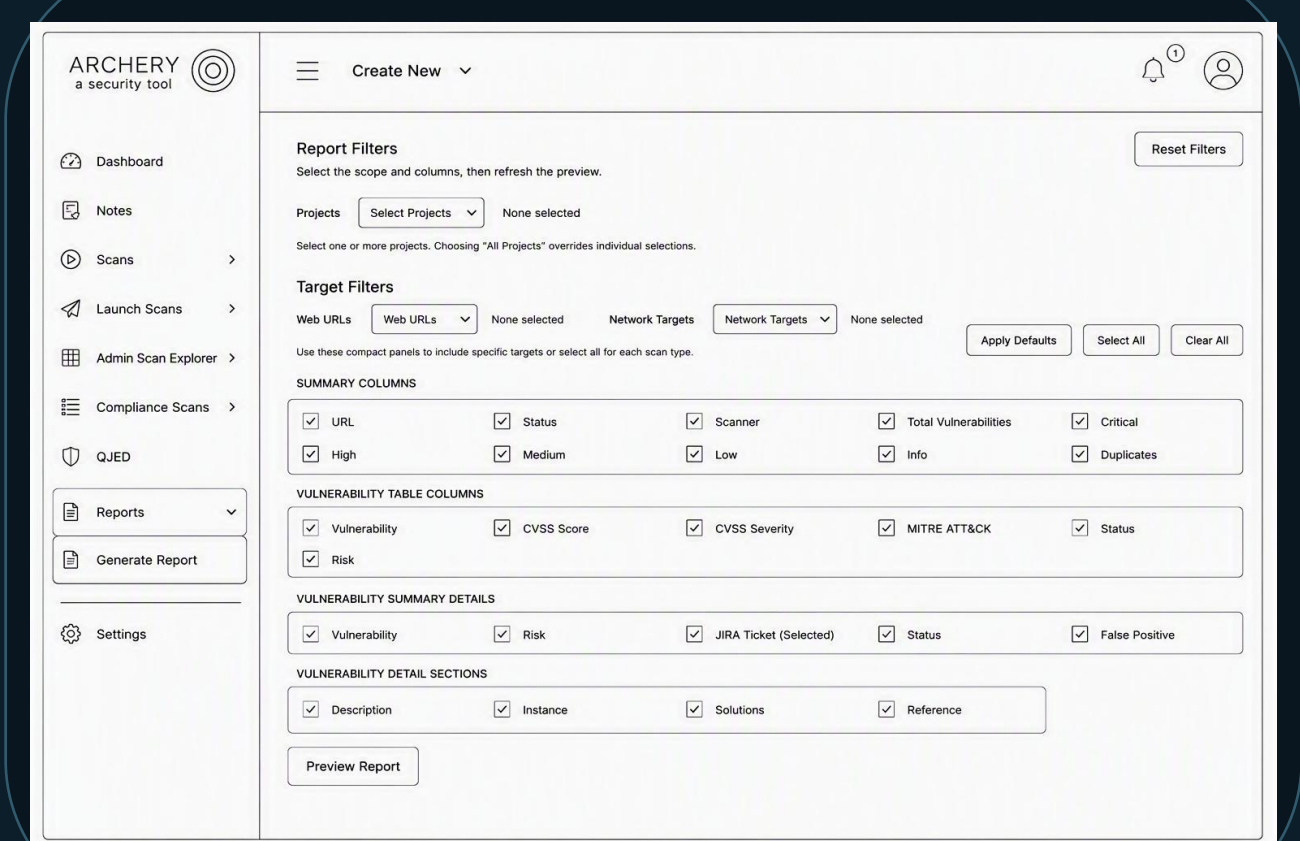
Unified Vulnerability Dashboard

Centralized scan summaries, severity distribution, trends and visibility across authorized projects.



Automated Scan Scheduling

Configure authorized targets, projects, execution time, recurrence and scanner settings.



Configurable Report Generation

Select report scope, fields and output options before previewing structured vulnerability reports.

SDG 9 CONTRIBUTION: Secure, centralized and extensible vulnerability management supports more resilient digital infrastructure.